Jacobian-lens trajectories under medical SFT

`gemma-3-270m-it`, `gemma-3-1b-it`, `gemma-3-4b-it`

July 30, 2026

Full fine-tune on `mix_default`, lr $1\mathrm{e}{-5}$, log-spaced checkpoints to step 4882

Setup

- **Three arms**, identical seed and data order: `gemma-3-{270m,1b,4b}-it`, full FT, lr $1\mathrm{e}{-5}$, completion-only loss.
- **Log-spaced checkpoints** $\{1, 2, 4, \ldots, 4096\}$ + final (4882). $t = 0$ is the untouched base, prepended by the probe.
- **Lens refit per checkpoint** on a frozen 1000-sequence generic (non-medical) WikiText corpus, taken from `jlens`'s own `load_wikitext_prompts` — the lens is a function of the weights.
- **Concordance** on the frozen 193-prompt probe set: 50 `general` + 50 `medical` (hand-authored one-hop cloze, style-matched control pairs) + 93 `multihop` from `jlens`'s `lens-eval-multihop.json`.
- **Behavioral** trajectory in parallel: MedQA, MedMCQA, PubMedQA, reported as *answered* accuracy with answer rate alongside. **Scored by an LLM judge**, not a regex — see the instrument slide.

All lens numbers below are the **mean over the last** $n/5$ **source layers** (the probe's `kl_final` / `top1_final`): 3 layers at 270m, 5 at 1b, 6 at 4b. Where the single final layer disagrees, it is called out explicitly — see the read-out caveat. Coverage: 14 / 15 / 14 points (270m / 1b / 4b) — all three complete. **Behavioral: 4b is judge-scored and complete (14 points incl.** t=0**); 270m and 1b are still on the superseded regex scorer** and are *not* comparable to it.

The probe step in detail

Runs in `.venv-jlens` (`transformers` 5.14.1 + `jlens @ 581d398`) — a second env, because the training pin 4.53.2 is hard-incompatible. Imports nothing from `gemma_med`.

Per checkpoint (base prepended as step 0, then `checkpoint-*`):

1. **Load** → `jlens.from_hf(hf, tok)`. Forced text-only `Gemma3ForCausalLM`; any loader leaving a weight missing is *rejected*, not accepted.
2. **Fit** a fresh lens: `jlens.fit(model, corpus, dim_batch=16, max_seq_len=128)` on the frozen 1000-seq WikiText corpus. *Dominates runtime.*
3. **Concordance**: for each of the 193 probes, `lens.apply(model, prompt, positions=[-1])` → per source layer, top-1 agreement and $\mathrm{KL}(\text{model} \,\|\, \text{lens})$ at the last token.
4. **Drift** vs. $t=0$, per layer: relative Frobenius $\|J_l - J_l^0\|/\|J_l^0\|$ and cosine.

Invariants that make the trajectory comparable

- **One load path for all points.** Letting `Auto*` pick per-path would read $t=0$ through the multimodal wrapper and later points through the text model — two forwards on one trajectory, silently corrupting drift.
- **Lens refit every time.** J is a function of the weights.
- **Corpus and probe set frozen** across all checkpoints and sizes.
- $t=0$ **lens cached in fp32**, so a resumed run's drift baseline matches a fresh run's exactly.

Output. One `fsync`'d JSON row per checkpoint in `metrics.jsonl`, plus W&B scalars keyed by training step. Resumable: completed steps are skipped and the fit checkpoints every 50 prompts, so a requeue does not wipe the trajectory.

Reading the metrics — the two drift measures

J_l = lens Jacobian at source layer l; J_l^0 = the same layer in the t=0 base lens. n = number of source layers (18 / 26 / 34 at 270m / 1b / 4b). Both measures are computed per layer, then averaged over all n; both are exactly 0 / 1 at t=0 by construction.

Metric	How it is computed	Direction
rel. Frobenius `relfro_mean`	$\|J_l - J_l^0\|_F \, / \, \|J_l^0\|_F$ — the *size* of the change, normalised by the base magnitude.	**Neither better nor worse.** It says *how far* the lens moved, not whether the move helped. Larger = more change.
cosine `cos_mean`	$\langle J_l, J_l^0 \rangle / (\|J_l\| \, \|J_l^0\|)$, flattened over all entries — the *direction* of the change.	**Neither.** 1 = pointing the same way as the base; lower = more rotation. Bounded above by 1.

Why both. They can move independently: a lens can grow in magnitude without rotating (relfro up, cosine ≈1) or rotate without changing size. In practice they track each other here, which is itself a mild sanity check on the fit.

Not comparable across sizes. J_l has a different shape at every arm, so only these normalised scalars travel — never a raw Jacobian entry.

Reading the metrics — the four faithfulness measures

All four compare the **lens read-out** against the **model's actual next-token distribution**, at the last prompt token, averaged over the prompts of one cue. They differ in which layers they read and how they score.

Metric	How it is computed	Direction
faithfulness KL `kl_final`	$\mathrm{KL}(\text{model} \,\|\, \text{lens})$, averaged over the **last** $n/5$ **layers**. *The workhorse metric.*	**Lower is better** — the lens predicts what the model actually says. $0 =$ perfect; unbounded above.
top-1 agreement `top1_final`	Fraction of prompts where $\arg\max$ lens $= \arg\max$ model, same last-$n/5$ layers.	**Higher is better**, range 0–1. Much coarser than KL — see caveats.
`top1_auc`	The same agreement averaged over **all** n layers, not just the top fifth. Despite the name, a mean rather than an integral.	**Higher is better** — agreement holds deeper down the stack, not only near the output.
`crossover_frac`	Shallowest layer whose top-1 agreement reaches 0.5, as a fraction of depth; 1.0 if it never does.	**Lower is better** — the lens locks onto the model's answer earlier in depth.

Cues: `general` / `medical` are the 50+50 hand-authored style-matched pairs; `multihop` the 93 externally sourced prompts. *The cue is the unit of comparison* — medical is read against general, never on its own. This deck reports `kl_final` and `top1_final`; the other two are computed and stored but not used.

Reading the metrics — direction, and what not to do

Behavioral metrics

- **answer rate** — % of generations that committed to an option. Higher better. A pure *format* measure.
- **answered accuracy** — accuracy among those only. Higher better. **This is the knowledge signal.**
- **raw accuracy** — non-answers counted wrong. Higher better, but confounded: it moves when *either* knowledge or formatting moves.

Why it matters: on 4b MedQA, raw accuracy falls 52.9 $\rightarrow$ 43.0 by step 512 — which reads as catastrophic forgetting. Answered accuracy over the same span is 53.1 $\rightarrow$ 51.4, essentially flat. The entire drop is the answer rate going 99.5% $\rightarrow$ 83.8%. Reporting raw alone inverts the finding, at both sizes where it has been checked.

Three rules for reading these numbers

1. **Never compare absolute KL across sizes.** `kl_final` averages a *different number of layers* per arm (3 / 5 / 6), over models of different depth and width. Only the **within-arm Δ from** t=0 is comparable across arms.
2. **Always read against t=0.** Every figure carries the untuned base as a dashed line or a pale bar. A finetuned-only number says nothing.
3. **Name the read-out.** "KL" is ambiguous: the layer mean and the single final layer disagree in sign on 270m medical. This deck is the **layer mean** throughout.

Directional summary. Drift (relfro $\uparrow$, cosine $\downarrow$) says the lens *changed*; KL $\uparrow$ and top-1 $\downarrow$ say the lens *got worse at predicting the model*. The two are independent: a lens can move a long way and stay faithful, or barely move and lose faithfulness. Most of this deck is about which of those happened at each size.

Five findings

1. **Drift has a sharp onset, then saturates** — and at 4b it partly *retreats*. The Jacobian finishes moving in the first $\sim$100 steps; the remaining 97% of training adds nothing.
2. **The faithfulness cost of medical SFT scales with model size.** ΔKL on `general` is $+0.09/+0.54/+2.08$ and on `multihop` $+0.02/+1.01/+1.63$ at 270m / 1b / 4b — monotone in size on both out-of-domain cues.
3. **The medical cue is not protected at 4b.** This overturns the two-arm reading. Medical KL goes $-0.05/-0.37/+\mathbf{1.19}$: flat at 270m, better at 1b, clearly *worse* at 4b.
4. **Medical SFT works at 4b — and the lens cost is paid anyway.** $+5.3/+8.3/+14.6$ pp answered accuracy on MedQA / MedMCQA / PubMedQA over the untuned base. **So the size where faithfulness degrades most is also the size that learns most**; the two are not in tension.
5. **Drift leads format collapse, both lead accuracy** — now confirmed at 4b. Drift vs. answer rate is $r=-0.74$ (MedQA) / $-\mathbf{0.91}$ (MedMCQA); drift vs. *answered* accuracy is $+0.10/+0.37$. The lens tracks formatting, not competence.

The two-arm version of this deck led with "medical holds or improves." At three sizes that is no longer the finding: what generalises is the *cost*, which grows with scale on every cue — while the *benefit* grows with scale too.

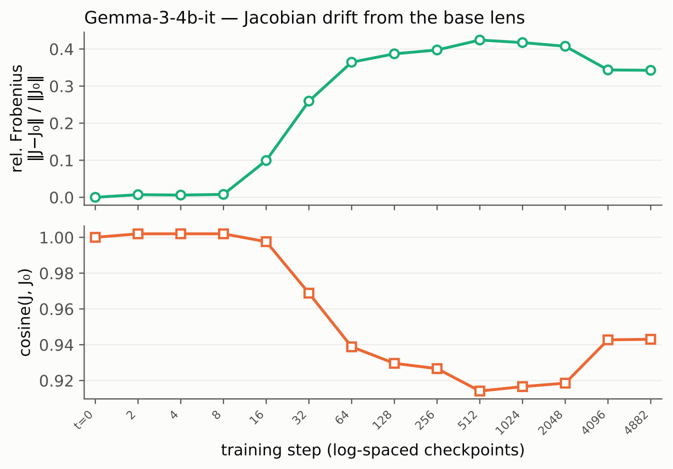
1. Drift onsets sharply, then saturates

	onset	plateau	peak relfro	final relfro / cos (layer mean)
`270m-it`	steps 32–64	$\sim$step 128	0.250	0.249 / 0.976
`1b-it`	steps 8–32	$\sim$step 64	0.308	0.279 / 0.956
`4b-it`	steps 8–32	$\sim$step 128	0.424	0.343 / 0.943

- Nothing moves for the first $\sim$8 steps (mean rel. Frobenius ≤ 0.008 at all three sizes).
- Then the Jacobian swings hard over roughly *one decade* of steps, and flatlines. **Onset is earlier and the swing larger the bigger the model** (final relfro $0.249 < 0.279 < 0.343$).
- **New at 4b: a late partial retreat.** Peak 0.424 at step 512 $\rightarrow$ 0.343 at 4882, a 19% walk-back toward the base lens. 1b shows a hint of the same (-9% from its step-2048 peak); 270m is flat.

Most robust result — holds at all three sizes, and under either the layer mean or the per-layer max (max: 0.349 / 0.398 / 0.479).

1. Drift — `4b-it`



The cosine panel reads slightly *above* 1.0 at steps 2–8. Cosine cannot exceed 1: this is fp32 dot-product accumulation over $\sim$6.5M entries per layer while the norms use a stabler path. It is visible only where drift is $\approx$0 and affects no step ≥ 16.

2. The faithfulness cost scales with model size

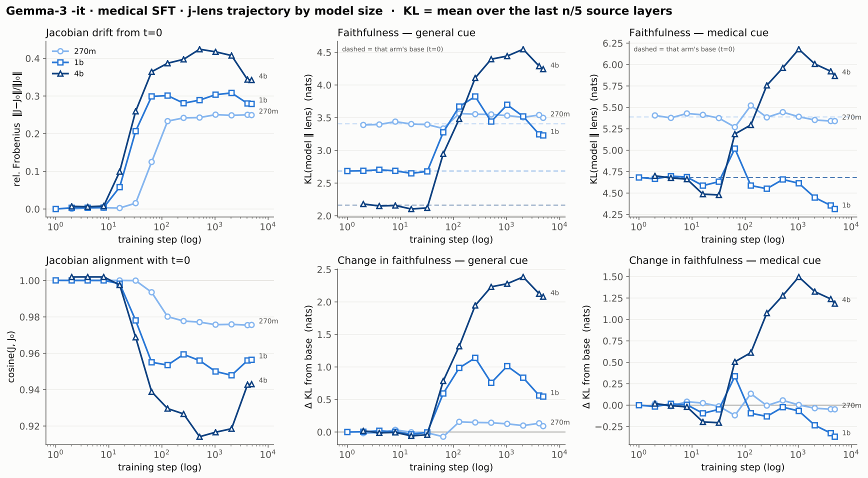
Lens$\leftrightarrow$model KL, mean over the last $n/5$ layers (the probe's `kl_final`) — *lower = lens predicts the model better*. Base $\to$ final checkpoint.

	KL `general`	KL `medical`	KL `multihop` (OOD)
`270m-it`	3.407 $\to$ 3.497 (+0.09)	5.388 $\to$ 5.340 (−0.05)	1.958 $\to$ 1.975 (+0.02)
`1b-it`	2.684 $\to$ 3.229 (+0.54)	4.682 $\to$ 4.314 (−0.37)	2.423 $\to$ 3.430 (+1.01)
`4b-it`	2.163 $\to$ 4.243 (+**2.08**)	4.683 $\to$ 5.867 (+**1.19**)	4.088 $\to$ 5.715 (+**1.63**)
top-1 `medical`:	270m 0.260 $\to$ 0.260 (flat)	1b 0.448 $\to$ 0.572	4b 0.677 $\to$ 0.660

- **Monotone in size on both out-of-domain cues.** `general` +0.09 $\to$ +0.54 $\to$ +2.08; `multihop` +0.02 $\to$ +1.01 $\to$ +1.63. Nothing here reverses with scale.
- **Bigger models start more faithful and lose more of it.** Base `general` KL falls with size (3.41 $\to$ 2.68 $\to$ 2.16) while the damage grows — so the ranking at $t=0$ is the reverse of the ranking after SFT.

Read-out check — the 270m disagreement does *not* recur. At 4b the layer mean and the single final layer agree in sign on all three cues (`general` +2.08 vs. +0.34, `medical` +1.19 vs. +0.09, `multihop` +1.63 vs. +0.24). The lone sign flip in this study remains 270m `medical`.

2. All three arms on one axis



Arms shaded light→dark by size (an *ordinal* ramp — size is ordered, so it does not take categorical hues). x is the real training step on a log axis, since the arms have different step grids; $t=0$ is each arm's dashed baseline (top row) and the zero line (bottom row), never a point.

3. The medical cue is not protected at 4b

The two-arm deck read this section as "medical holds or improves, and the cost is paid out-of-domain." At 4b that is false — medical degrades by $+1.19$ nats.

	ΔKL `medical`	reading
`270m-it`	-0.05	flat — the lens barely registers the fine-tune
`1b-it`	-0.37	genuinely better, and top-1 agrees ($+0.124$)
`4b-it`	$+1.19$	clearly worse, and top-1 agrees (-0.017)

- **What survives:** medical is always the *least damaged* cue of the three at its own size. At 4b, `general` $+2.08$ and `multihop` $+1.63$ are both worse than medical's $+1.19$.
- **What does not survive:** "medical improves." That was a 1b-shaped result, and 1b is the only arm where it happens.

So medical SFT does buy the medical cue *relative* ground at every size — but at 4b it buys relative ground while everything, including medicine, gets absolutely less faithful.

3. The domain gap narrows — at 4b from one end only

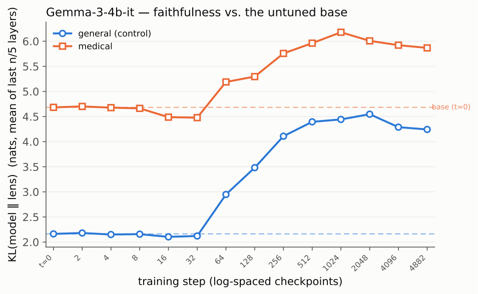
Within each arm the `medical` cue starts 2–2.5 nats less faithful than `general`. **All three arms narrow that gap** — but not by the same mechanism.

	t=0	final	Δ	closes by
`270m-it`	1.982	1.843	−0.139	both ends
`1b-it`	1.998	1.085	−**0.912**	both ends
`4b-it`	2.520	1.624	−**0.896**	general only

At 270m and 1b the gap closes **from both ends** — medical improves *and* general regresses toward it.

At 4b it closes from one end: both cues degrade, `general` just faster (+2.08 vs. +1.19). **An identical** −0.9 **gap change means opposite things at 1b and 4b** — which is why the gap is never reported on its own.

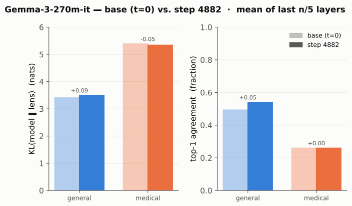
`multihop` degrades at *all three* sizes (+0.02 / +1.01 / +1.63).



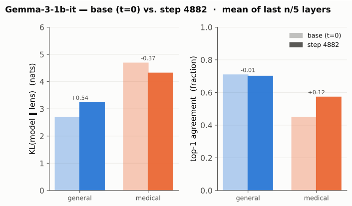
The 4b panel alone: *both* curves rise away from their dashed t=0 baselines, the general one further. The 270m and 1b equivalents are on the previous slide.

3. Base vs. finetuned, side by side

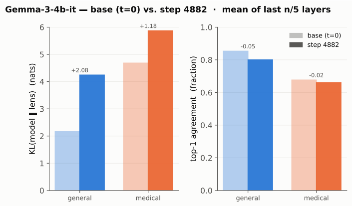
270m-it



1b-it



4b-it



Pale bar = untuned base (t=0), solid = step 4882. All three figures are pinned to the same axes (KL 0–6 nats, top-1 0–1), so the panels are directly superimposable — but compare the Δ *labels*, not the bar heights: absolute KL is not comparable across sizes (different depth and width, and a different number of layers in the $n/5$ mean). Read that way, 270m's largest KL move (+0.09) is about a quarter of 1b's smallest (−0.37), and 4b's smallest (+1.19) is larger than either arm's biggest.

4. The scoring instrument changed

Behavioral scoring is no longer a regex over the generations. It is an **LLM judge** (`gemma_med.judge`): the model's response, the options and the gold answer go to `claude-haiku-4-5` under a schema-constrained verdict of `correct` / `incorrect` / `no_answer`.

Why it is a better instrument

- It matches option *content*, not the letter: "A. *[text of option C]*" scores as a choice of C. The regex scored it as A.
- It is told the gold answer and **forbidden to re-derive it**, so the metric stays benchmark accuracy — not agreement between two models.
- `no_answer` is the successor to "unparsed", so the three-way split survives intact.

What it costs you

- Judge numbers are not comparable to regex numbers. It is strictly more generous, so a rise could be the instrument. **Only the 4b arm has been re-scored**; the 270m and 1b tables that follow are still regex.
- The judge is **frozen** per trajectory and recorded in `judged_summary.json`, exactly like the fit corpus and probe set.

Cross-validated. On 300 identical MedQA rows at 4b step-1024, `claude-haiku-4-5` and the lab's self-hosted Kimi agree to within 1.3 pp raw (45.0 vs. 46.3) and 0.3 pp on answer rate. Freezing the judge is about reproducibility, not about either being wrong.

4. Behavior — `4b-it` learns, and learns most

Judge-scored. Answered accuracy, with answer rate in parentheses. t=0 is the untuned `gemma-3-4b-it`.

	MedQA		MedMCQA		PubMedQA	
`4b-it` t=0	53.1	(99.5%)	46.8	(99.7%)	58.2	(100%)
`4b-it` step 512	51.4	(83.8%)	46.2	(89.2%)	73.7	(99.6%)
`4b-it` final	**58.4**	(96.5%)	**55.1**	(92.2%)	**72.8**	(99.4%)
Δ vs. base	+5.3		+8.3		+14.6	

- **The gain arrives late.** Nothing moves on MedQA/MedMCQA until step ∼2048; the first two thirds of training only degrade format.
- **PubMedQA is the exception** — it climbs from step 16 and is most of the way up by step 512, well before the MCQ benchmarks move.
- Mid-trajectory raw accuracy is a trap. Raw MedQA bottoms at 43.0 (step 512) while answered accuracy is 51.4 — see the metrics slide.

Not comparable to the 270m/1b table overleaf: that one is regex-scored on the superseded instrument.

4. Behavior — `270m-it` / `1b-it` (regex-scored, superseded instrument)

Parsed accuracy (parse rate in parentheses). Retained from the previous version of this deck. These have not been re-scored with the judge and must not be tabulated against the 4b numbers.

	MedQA		MedMCQA		PubMedQA	
`270m-it` $t=0$	24.6	(69%)	27.2	(13%)	54.5	(7%)
`270m-it` final	25.5	(49%)	25.0	(49%)	48.0	(5%)
`1b-it` $t=0$	31.9	(100%)	32.6	(100%)	52.6	(100%)
`1b-it` final	**38.5**	(25%)	**37.8**	(75%)	**58.1**	(97%)

- **270m learns nothing.** MedQA/MedMCQA sit near chance throughout; every apparent movement is format churn (MedMCQA parse 13%→49%; PubMedQA swings 5%→88%→5%).
- **1b gains** +6.6 / +5.2 / +5.5 **pp** parsed — while format compliance collapses, worst on MedQA (100% → 25%).

The *qualitative* 270m-vs-1b split (one learns, one only churns format) is instrument-independent. The pp values are not; re-judging both arms is the top item on Next.

4. A second failure mode: collapse onto the majority label

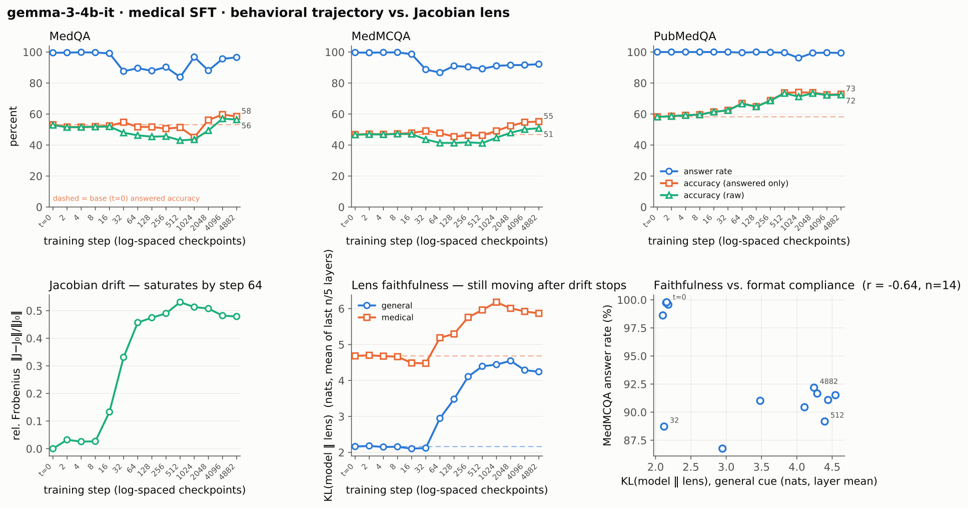
Format collapse is not the only thing that goes wrong mid-training. Share of *answered* items assigned to the dominant label, against that label's prevalence in the gold answers:

	MedQA ("A")		MedMCQA ("A")		PubMedQA ("yes")	
	chosen	gold	chosen	gold	chosen	gold
$t{=}0$	21.9	27.7	22.2	32.2	51.2	55.2
step 512	31.4	27.7	41.9	32.2	64.1	55.2
step 1024	**57.2**	27.7	**52.5**	32.2	69.9	55.2
final	34.3	27.7	38.5	32.2	65.8	55.2

- **The base model is *anti*-majority** (21.9 vs. 27.7). SFT pushes it through neutral and overshoots.
- **Step 1024 is a spike, not a trend** — it recovers. Worth knowing before building a story on any single checkpoint.
- **PubMedQA is the control that reframes it.** It has no "A" to collapse onto, yet shows the same overshoot on "yes". So this is **majority-label over-commitment, not position bias.**

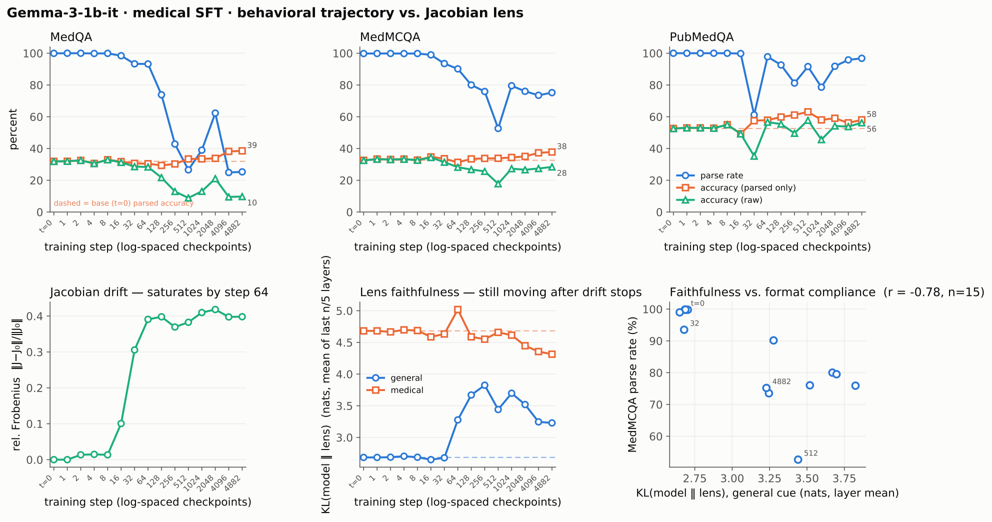
The two modes trade off: steps 32–512 lose format compliance while answered accuracy holds; step 1024 recovers compliance but concentrates its answers; 2048+ resolves both.

4. Joined behavioral + lens trajectory — `4b-it`

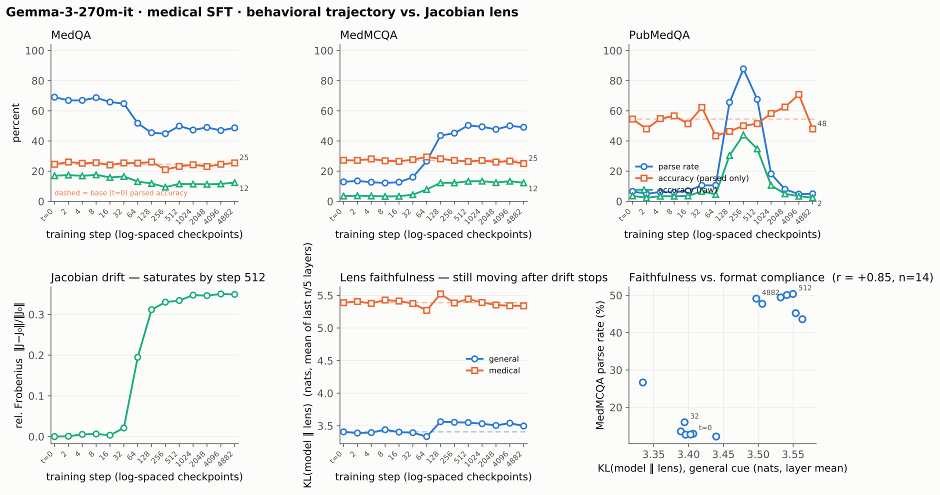


Judge-scored. Top row: answer rate (blue) separates cleanly from answered accuracy (orange) from step 32 — the format/knowledge split made visible.

4. Joined behavioral + lens trajectory — `1b-it`



4. Joined behavioral + lens trajectory — `270m-it`



The KL panel is the layer mean, as everywhere else in this deck; at the single final layer the medical curve rises instead.

5. Timing: drift leads format collapse, both lead accuracy

The same ordering holds at both sizes where behavior exists. `4b-it` (judge-scored):

Signal	Onset	Notes
Jacobian drift	steps 16–32	0.027 $\rightarrow$ 0.331; plateaus by step 64
Format compliance	step 32	MedQA answer rate 99 $\rightarrow$ 88%, floor 83.8% at 512
Task accuracy	steps 2048+	MedQA answered 51 at step 512; 58–60 only at 4096–4882

And the correlations say which one the lens is tracking:

Pearson r vs. drift	MedQA	MedMCQA	PubMedQA
answer rate (format)	−0.737	**−0.912**	−0.415
answered accuracy (knowledge)	+0.100	+0.373	+0.910

Interpretation

The Jacobian has **finished moving before the model has learned any medicine**, and it moves *with* format collapse ($r \approx -0.9$ on MedMCQA), not with competence ($r \approx +0.1$ on MedQA). Whatever the lens tracks here is the format / style / alignment shift. *PubMedQA is the exception — its accuracy rises early, on the same clock as drift.*

Caveats — the measurements

- **Top-1 is coarse.** n=50 prompts for `general/medical` (93 for `multihop`), so *per-layer* top-1 quantizes at 0.02; the reported statistic averages 3 / 5 / 6 layers, giving a 1/150–1/300 grain that makes small moves look more precise than they are. Single-checkpoint wiggles of $\pm$0.04 are still noise. *Trust the KL trends, not top-1.*
- **Read-out choice is not innocent.** Layer-mean and final-layer KL disagree in sign on 270m medical. Every claim here uses the layer mean; an earlier draft of this deck used the final layer and reported a 270m$\leftrightarrow$1b *inversion* that does not survive. Report the statistic, not just the number. (It does not recur at 4b.)
- **A numerical artifact in the drift cosine.** At 4b steps 2–8, `cos_mean` reads 1.001–1.003; cosine cannot exceed 1. The fp32 `dot` over $\sim$6.5M entries per layer is biased against the stabler norm computation. Only visible where drift $\approx$0; no reported claim depends on it, but `jac_drift` should accumulate in fp64.
- **The `general/medical` cues are hand-authored**, not sampled from a corpus — the domain-gap result inherits whatever bias is in those 100 prompts. `multihop` is the one cue with an external, independently-constructed source.

Caveats — the scope

- **Three sizes is a thin scaling trend.** The monotone cost on `general/multihop` now rests on three points spanning $\sim 15\times$ in parameters, which is better than the two-point version it replaces — but 270m may simply be too weak to learn the task at all, so the low end is a weak anchor. 12b/27b would settle it.
- **The 4b behavioral arm now exists** — findings 4 and 5 hold at the size where the lens result is strongest. The gap moved rather than closed: 4b is judge-scored and 270m/1b are regex-scored, so **no cross-size behavioral comparison is currently legitimate**. Re-judging the two smaller arms ($\sim$\$32 and $\sim$22 min each) is what unblocks it.
- **One hyperparameter setting.** All three arms are full FT at lr $1e{-5}$ on one mixture. Whether the size trend is really about *size*, or about a fixed LR being progressively too large as the model grows, is not separable from these runs alone.
- **Coverage gaps.** The 4b run has no `checkpoint-1`, so its grid starts at step 2 (270m likewise; only 1b probed step 1). No `-pt` arm exists at any size — the pt-vs-it starting-point axis is entirely unstarted.

Next

1. **Re-judge 270m and 1b** with the same frozen judge — now the most valuable missing piece. Until then the deck has one judge-scored arm and two regex-scored ones, and *no* behavioral claim can be made across sizes. One command per arm, ∼\$32 and ∼22 min each.
2. **12b (LoRA, caveated)** to test whether the monotone cost keeps climbing or turns over. Three points is a direction, not a curve.
3. **Widen the `general`/`medical` cues** beyond the 100 hand-authored prompts, to check the domain-gap collapse is not an artifact of how those pairs were written.
4. **Open a `-pt` arm** at 1b: does a non-aligned starting point show the same drift-before-competence ordering?
5. **Anchor against `medgemma-27b-text-it`** on the same probe set to place the trajectory endpoints on an absolute scale.

Answered since the last version: the **4b behavioral trajectory now exists** (judge-scored, 14 points incl. t=0) — medical SFT gains $+5.3/+8.3/+14.6$ pp at 4b, and the drift-leads-format ordering is confirmed at $r = -0.91$. Also: the lens trajectory exists at 4b, and the layer-mean/final-layer disagreement does *not* recur there.

Data provenance

Artifact	Path
Lens metrics (270m)	`eval/jlens/270m_it_full/metrics.jsonl` (14 rows)
Lens metrics (1b)	`eval/jlens/1b_it_full_v3/metrics.jsonl` (15 rows)
Lens metrics (4b)	`eval/jlens/4b_it_full/metrics.jsonl` (14 rows)
Joined traj. (270m)	`eval/traj/270m_it_full/joined.json` (regex)
Joined traj. (1b)	`eval/traj/1b_it_full_v3/joined.json` (regex)
Joined traj. (4b)	`eval/traj/4b_it_full/joined.json` (judge)
Fit corpus	`data/jlens/fit_corpus_v2.txt` (1000 WikiText seq, frozen)
Probe set	`data/jlens/probe_set_v2.tsv` (50/50/93, frozen)
Figures	`figs/{270m_it_full,1b_it_full_v3,4b_it_full,arms}/`
Probe jobs	25776556 (1b), 25799677 (270m), 25948088–25948100 (4b, 13-way fanout)

All three runs logged `fit_corpus=1000 probes=193` — i.e. the current frozen v2 inputs. The v1 files (`fit_corpus.txt`, 40 seq; `probe_set.tsv`, 30 prompts) are legacy and still present in `data/jlens/`; they were used only by the superseded runs.

4b was probed one Slurm job per checkpoint ($\sim$3.5 h each, $\sim$7 h wall clock vs. $\sim$50 h serial), all 13 workers reusing the *same* fp32 $t=0$ lens so drift shares one baseline; per-worker rows merged by `scripts/merge_jlens_fanout.py`, which refuses the merge if any non-base step appears twice. `4b_it_validate{,2}` are the earlier 5-point wrapper-validation spikes (base + steps 1–8), superseded.

Superseded: `1b_it_full_v2` (v1 30-prompt set, no `multihop`), `1b_it_full_probev2` (v2 inputs, immediately re-run as v3), `270m_verify` (Phase-0 8-point smoke test, v1 inputs).

Data provenance — the behavioral judge (4b)

Field	Value
Scorer	`gemma_med.judge` (replaces `answer_parsing.py`)
Provider	`anthropic`, Message Batches API (half price)
Judge model	`claude-haiku-4-5` **frozen for the trajectory**
Verdicts	`correct` / `incorrect` / `no_answer`, schema-constrained
Volume	83,384 items over 42 batches, $\sim$11 min, $\sim$\$34
Judgments	`eval/traj/4b_it_full/step-*/*_judgments.jsonl`
Provenance	`.../step-*/judged_summary.json` (13 files)
t=0 dir	`eval/gemma-3-4b-it-baseline/`

`judge_self_disagreement` — **a quality signal that scores nothing.** It counts rows where the judge's verdict string conflicts with the option it actually picked: 0.2–0.5% at the ends of the trajectory, peaking at 2.8% (step-64 MedMCQA, 117/4183).

It changes no number. `correct` and `answered` are both recomputed from `chosen`, never from `verdict`. Of those 117 rows $\sim$73% are one benign shape: the model emits long "Step 1: Restating the question..." CoT and hits the generation limit before naming an option, so the judge sets `chosen=none` (correct) but labels the verdict `incorrect` rather than `no_answer`. The statistic tracks **how rambly the judged model is**, not how confused the judge is — which is why it peaks in the format-collapse region and subsides at both ends.